

Navigating the Safety Landscape: Measuring Risks in Finetuning Large Language Models

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Abstract

Safety alignment is crucial to ensure that large language models (LLMs) behave in ways that align with human preferences and prevent harmful actions during inference. However, recent studies show that the alignment can be easily compromised through finetuning with only a few adversarially designed training examples. We aim to measure the risks in finetuning LLMs through navigating the LLM safety landscape. We discover a new phenomenon observed universally in the model parameter space of popular open-source LLMs, termed as “safety basin”: random perturbations to model weights maintain the safety level of the original aligned model within its local neighborhood. However, outside this local region, safety is fully compromised, exhibiting a sharp, step-like drop. This safety basin contrasts sharply with the LLM capability landscape, where model performance peaks at the origin and gradually declines as random perturbation increases. Our discovery inspires us to propose the new VISAGE safety metric that measures the safety in LLM finetuning by probing its safety landscape. Visualizing the safety landscape of the aligned model enables us to understand how finetuning compromises safety by dragging the model away from the safety basin. The LLM safety landscape also highlights the system prompt’s critical role in protecting a model, and that such protection transfers to its perturbed variants within the safety basin. These observations from our safety landscape research provide new insights for future work on LLM safety community. Our code is publicly available at <https://github.com/ShengYun-Peng/llm-landscape>.

1 Introduction

Safety alignment is the foundation to bring LLMs’ behaviors in line with human preferences and restrict harmful behaviors at inference time [43, 44, 3]. Though aligned LLMs have adopted one or a combination of the safety alignment methods, *e.g.*, reinforcement learning from human feedback (RLHF) [35], instruction tuning [46], direct preference optimization (DPO) [39], and rejection sampling [33], LLM safety can easily be compromised by finetuning with only a few adversarially designed training examples. For instance, both GPT-3.5 Turbo and LLaMA-2 [44] fail to refuse users’ harmful queries after only finetuning with 10-shot harmful examples [37]. This brings practical safety concern to model deployment as customization is the desirable way for specific use case. In this paper, we explore the following fundamental problems in LLM safety: **Are all open-source LLMs equally vulnerable to finetuning? Why can simple finetuning easily break LLM’s safety alignment? How fast does the model start to break during finetuning?**

We discovered that all these questions can be addressed by navigating the LLM safety landscape. In deep learning literature, visualization of the model landscape has significantly improved our comprehension of generalization errors, optimization trajectories, model ensembles, and adversarial

A. Safety basin universally appears in open-source LLMs’ parameter spaces. Randomly perturbing model weights maintains safety level of original aligned model (light purple dot) in its local neighborhood.

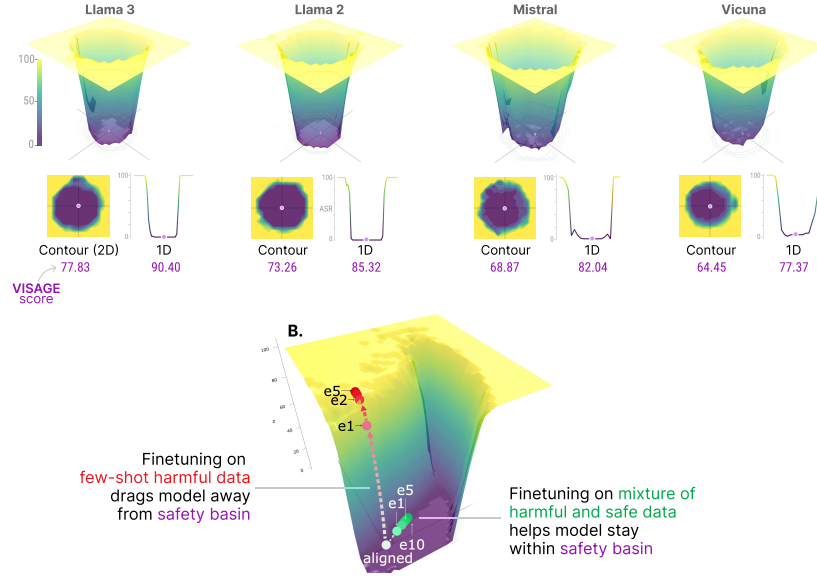


Figure 1: **A.** “Safety basin”, a new phenomenon observed universally in the model parameter space of popular open-source LLMs. Our discovery inspires us to propose the new VISAGE safety metric that measures the safety in LLM finetuning by probing its safety landscape. **B.** Visualizing the safety landscape of the aligned model also enables us to understand why finetuning with harmful data compromises safety but finetuning with both harmful and safe data preserves the safety.

robustness [31, 30, 13, 50, 42]. In this paper, we introduce the notion of LLM safety landscape and quantify the risk in finetuning LLM by exploring different directions of perturbing model weights. When provided with a single model, we sample a random normalized direction to visualize its local variations. When given two models varied by fine-tuning, we utilize linear interpolation to visualize the changes between them. The shape of the landscape dictates the fine-tuning attributes: a sharp change in the safety metric indicates that the aligned model is a local minimum, making it challenging to find a point that is both safe and useful, whereas a flat local landscape offers more opportunities to discover a model that better balances safety and usefulness. Our landscape navigation provides a suite of four new insights that facilitate the understanding the LLM safety (Fig. 1):

1. **We discover a new phenomenon observed universally in the model parameter space of popular open-source LLMs, termed as “safety basin”: random perturbations to model weights maintain the safety level of the original aligned model within its local neighborhood. However, outside this local region, safety is fully compromised, exhibiting a sharp, step-like drop.** The safety basin is evident in both 1D and 2D safety landscape of LLaMA2, LLaMA3, Vicuna, and Mistral across various random directions and different safety benchmarks. This safety landscape contrasts sharply with the LLM capability landscape, where model performance peaks at the origin and gradually declines as random perturbation increases. Our discovery inspires us to propose the new VISAGE safety metric, the acronym for volumetric index for safety alignment guided by explanation, which measures the safety of an LLM’s local region in model parameter spaces. (Sec. 3)
2. **Visualizing the safety landscape of the aligned model enables us to understand, for the first time, how finetuning compromises safety by dragging the model away from the safety basin.** We discover that different LLMs have varying rates of vulnerability to finetuning, and our task agnostic VISAGE safety metric measures the risks in finetuning without assumptions on the finetuning dataset, where a higher VISAGE score means the model after finetuning is safer. Though finetuning can easily break the safety alignment, we demonstrate that as long as the finetuning process stays within the safety basin, the safety of the finetuned model remains intact. (Sec. 4)
3. **LLM safety landscape also highlights the system prompt’s critical role in protecting a model, and that such protection transfers to its perturbed variants within the safety basin.** We evaluate the impact of system design on LLaMA2, LLaMA3, Vicuna, and Mistral, using each

LLM’s default system prompt as the baseline. From an attacker’s standpoint, we find that both removing the default system prompt and using simple roleplaying jeopardize the safety alignment, with the former exhibiting greater potency. From a defender’s perspective, we discover that LLaMA2’s original system prompt universally enhances safety across models, and safety prompts optimized through prompt tuning for a specific model also enhances safety for all models inside the safety basin. (Sec. 5)

4. **When evaluating the safety landscape using jailbreaking queries, we find that these queries are highly sensitive to perturbations in model weights.** We have collected the adversarial prompts targeting LLaMA2 and Vicuna, generated by jailbreak attacks from the literature [53, 8, 9]. Our safety landscape analysis shows that although the aligned model is vulnerable to jailbreak attacks, slightly perturbing the model weights in the local space of the aligned model can significantly lower the attack success rate (ASR) of these jailbreaking attacks. A naive defense method is to perturb the model weights before generating the response. However, attackers can also create stronger attacks that target both the aligned model and multiple perturbed models in its local region. These observations from our safety landscape research provide new insights for future work on LLM attacks and defenses. (Sec. 6)

2 Background and Related Works

LLM safety alignment. LLMs are language models with a large number of parameters trained on web-scale text corpora [5, 1, 44, 10, 27]. LLMs have exhibited emergent capabilities that can be broadly applied in a task-agnostic manner, such as in-context learning [5], chain-of-thought reasoning [47], and mathematical reasoning [25]. These capabilities are largely attributed to aligning LLMs with expected human values and intentions, which involves training the model to follow instructions and being helpful, truthful, and harmless [35, 24, 40]. Specifically, harmless is achieved by safety alignment that empowers the LLM with safety guardrails so that the model can refuse harmful instructions. Common safety alignment techniques are instruction tuning [46], RLHF [35], DPO [39], rejection sampling [33], and self-alignment [41]. However, these techniques are not designed to cover the safety risks that may arise from the subsequent custom finetuning and jailbreak attacks. Recent work has shown that both simple finetuning [37] and jailbreak attacks [8, 53] can circumvent safety guardrails of aligned LLMs.

LLM harmful finetuning attacks and defenses. Finetuning is widely employed to customize open-source LLMs for downstream applications [18, 11]. Typically, finetuning directly updates the parameters of pretrained models using a small dataset to enhance performance on downstream tasks. However, finetuning with a few adversarially designed training examples, or even with a benign dataset, can compromise the safety alignment of LLMs [48]. Qi et al. [37] finetuned GPT-3.5 Turbo and LLaMA2-7b-chat with only 10 harmful examples, but the safety guardrails were undermined in both LLMs. Zhan et al. [49] removed the safety protections of GPT-4 with 95% success with only 340 examples trained with the OpenAI’s finetuning API. A new line of research aims to defend against such harmful finetuning attacks at both the alignment stage and the user finetuning stage, *e.g.*, Vaccine [23], Lisa [22], Antidote [20], Booster [21], and targeted Vaccine [32]. Safety-aware LLM fine-tuning, such as Safe LoRA [19], can mitigate safety degradation after fine-tuning by steering the model updates toward the direction of better alignment. In this paper, we probe into the mechanism of the harmful finetuning attack via navigating the safety landscape.

Enhancing alignment with safety prompts. To communicate with LLM with precise and task-specific instructions, Bsharat et al. [6] presented a comprehensive principled instructions and guidelines to improve the quality of prompts for LLMs. Recently, safety researchers have also experimented with various prompts to either break or enhance LLM safety alignment. On the attack side, Jin et al. [28] used roleplaying to automatically and iteratively generate harmful prompts. On the defense side, Zheng et al. [51] leveraged prompt tuning to enhance LLM safety by directly optimizing the vanilla system prompt into safety prompt. Safety prompt is a method of safeguarding LLMs against harmful queries without changing the model weights; they are prepended to the user input or serve as a system prompt.

Jailbreaking aligned LLMs with adversarial attacks. A class of vulnerabilities known as “jailbreaks” has recently been shown to cause LLMs to violate their alignment safeguards [7, 38, 45]. Jailbreaks based on human prompt strategies rely on deception and social engineering to elicit objectionable content from LLMs, requiring creativity, manual dataset curation, and significant human